Presenting Regression Results Using R

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Today's Goal

Learn how to present regression results professionally using R

Why this matters: - Running a regression presenting it well - Good presentation = better understanding - Different formats for different audiences

What is Regression?

The Big Picture

Regression answers: **How does X affect Y?**

Example: - Does **weight** affect **fuel efficiency**? - Does **experience** affect **salary**? - Does **price** affect **sales**?

The Simple Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

- Y = outcome (what we want to predict)
- X = predictor (what influences the outcome)
- β = effect size (how much X affects Y)

Regression in 60 Seconds

The Code

```r
# Step 1: Load data
data(mtcars)

# Step 2: Fit model
model <- lm(mpg ~ wt + cyl, data = mtcars)

# Step 3: See results
summary(model)
```

What the Results Mean

- **Coefficient for `wt`** = -3.2: Each additional 1000 lbs → -3.2 mpg (fuel gets worse)
- **P-value** = 0.0001: This effect is real (not by chance)
- **R^2 = 0.83**: The model explains 83% of fuel efficiency variation

Presenting Results: Option 1 - Tables

Publication Table

```r
library(stargazer)
library(tidyverse)

data(mtcars)
model1 <- lm(mpg ~ wt + cyl, data = mtcars)
model2 <- lm(mpg ~ wt + cyl + hp, data = mtcars)

stargazer(model1, model2, type = "text")
```

```
=================================================================
                                   Dependent variable:
                    ---------------------------------------------
                                           mpg
```

```
                                 (1)                     (2)
-----------------------------------------------------------------
wt                            -3.191***               -3.167***
                               (0.757)                 (0.741)

cyl                           -1.508***                -0.942*
                               (0.415)                 (0.551)

hp                                                     -0.018
                                                       (0.012)

Constant                      39.686***               38.752***
                               (1.715)                 (1.787)

-----------------------------------------------------------------
Observations                     32                      32
R2                              0.830                   0.843
Adjusted R2                     0.819                   0.826
Residual Std. Error        2.568 (df = 29)         2.512 (df = 28)
F Statistic           70.908*** (df = 2; 29) 50.171*** (df = 3; 28)
=================================================================
Note:                                 *p<0.1; **p<0.05; ***p<0.01
```

When to use: Academic papers, formal reports

Presenting Results: Option 2 - Charts

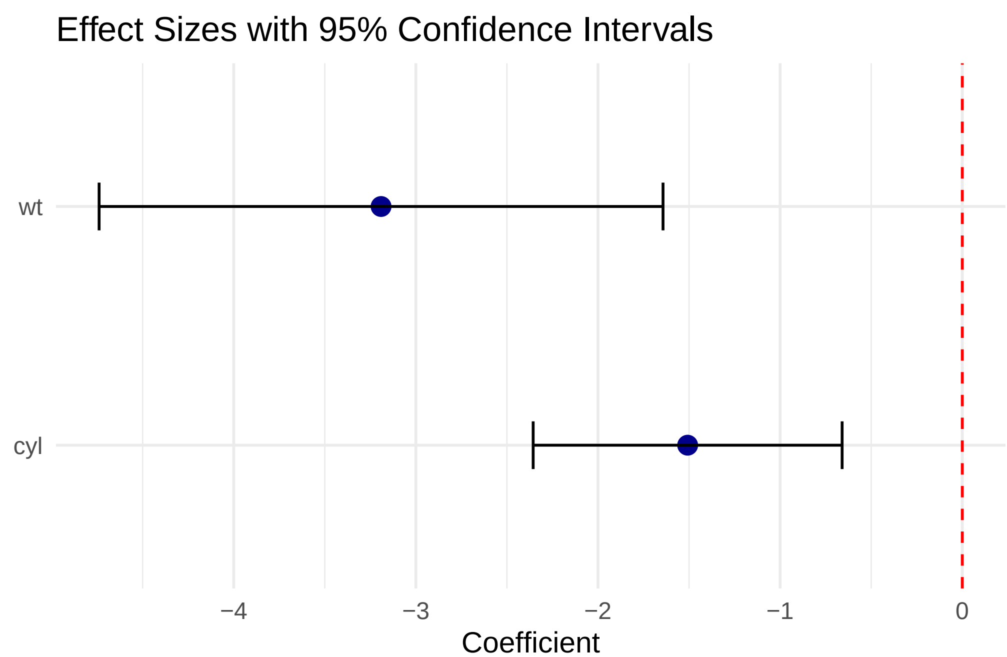
Coefficient Plot

```r
library(broom)

results <- tidy(model1, conf.int = TRUE)

ggplot(results %>% filter(term != "(Intercept)"),
       aes(y = term, x = estimate)) +
  geom_point(size = 3, color = "darkblue") +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high),
                width = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Effect Sizes with 95% Confidence Intervals",
```

```
      x = "Coefficient", y = "") +
  theme_minimal()
```



When to use: Presentations to non-technical audiences

Presenting Results: Option 3 - Predictions

Show Real-World Impact

```
scenarios <- tibble(wt = c(2.5, 3, 3.5, 4), cyl = 6)
pred <- predict(model1, newdata = scenarios, se.fit = TRUE)

scenarios %>%
  mutate(
    mpg = round(pred$fit, 1),
    ci = paste0("[", round(pred$fit - 1.96*pred$se.fit, 1),
                ", ", round(pred$fit + 1.96*pred$se.fit, 1), "]")
  ) %>%
  select(wt, cyl, mpg, ci) %>%
  knitr::kable(
```

```
  col.names = c("Weight (1000 lbs)", "Cylinders", "Predicted MPG", "95% CI")
)
```

Weight (1000 lbs)	Cylinders	Predicted MPG	95% CI
2.5	6	22.7	[21.4, 24]
3.0	6	21.1	[20.1, 22]
3.5	6	19.5	[18.4, 20.5]
4.0	6	17.9	[16.3, 19.4]

When to use: Business presentations, stakeholders

Best Practices

1. Know Your Audience

Audience	Format	Focus
Statisticians	Tables, detailed statistics	Coefficients, p-values, R^2
Business	Charts, scenarios	Real-world impact, predictions
General public	Simple visuals	"What does this mean for me?"

2. Always Show Uncertainty

Don't just say: "The effect is 3.2"

Say: "The effect is 3.2 (95% CI: 2.1 to 4.3)"

3. Validate Assumptions

Before presenting, check: - Linearity (is the relationship linear?) - Independence (are observations independent?) - Normality (are residuals normally distributed?)

4. Interpret in Context

Bad: "The coefficient is -3.2, p = 0.0001"

Good: "Each additional 1000 lbs of weight reduces fuel efficiency by 3.2 miles per gallon, a highly significant effect"

5. Avoid Common Mistakes

- Correlation = Causation
- Forgetting to mention sample size
- Ignoring outliers
- Over-interpreting small effects

Your Turn: Student Exercise

Exercise (10 minutes)

```r
# Load data
data(mtcars)

# Task 1: Fit a regression model
# Predict: mpg ~ hp (horsepower)
model <- lm(mpg ~ hp, data = mtcars)

# Task 2: Get summary
summary(model)

# Task 3: Interpret
# Q1: Does horsepower affect fuel efficiency?
# Q2: What is the effect size?
# Q3: Is it statistically significant?
```

You have 10 minutes. Go!

Wrap-Up

Key Takeaways

1. **Regression** = understand relationships between variables
2. **Tables** = detailed, formal
3. **Charts** = intuitive, accessible
4. **Predictions** = real-world impact
5. **Context** = numbers don't speak for themselves

Thank You!

Questions?